

Automated Parallel Goal-Based Adaptivity



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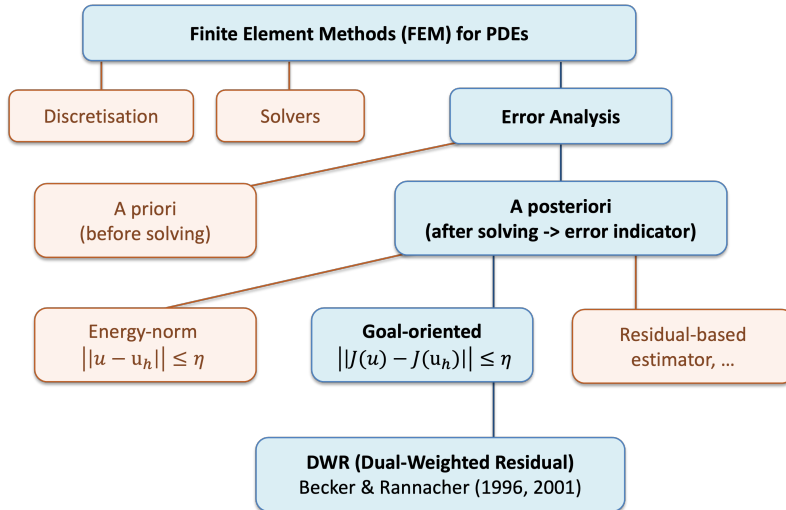
Motivation

Dual-Weighted Residual Method

Goal-based Adaptivity

Implementation Plan

Motivation: Error Analysis



DWR: Problem Formulation

- **Primal problem:** Find the exact solution $u \in V$ such that

$$F(u; v) = 0 \quad \forall v \in V.$$

- **Finite Element discretisation:** Find the discrete approximation $u_h \in V_h$ s.t.

$$F(u_h; v_h) = 0 \quad \forall v_h \in V_h.$$

- **Goal functional:** In practice, we are interested in a specific quantity of interest

$$J: V \rightarrow \mathbb{R}.$$

- **Objective:** Estimate the goal error

$$\eta \approx |J(u) - J(u_h)|,$$

and drive adaptivity so that it stays below a prescribed tolerance $\eta < \text{TOL}$.

DWR: Adjoint Problem and Error Identity

- **Adjoint problem:** To measure how the residual influences $J(u)$, introduce $z \in V$ such that

$$F'(u_h; \phi, z) = J'(u_h; \phi) \quad \forall \phi \in V.$$

- **Error identity:** For arbitrary approximations $\tilde{u} \in U$ and $\tilde{z} \in V$,

$$J(u) - J(\tilde{u}) = \frac{1}{2} \left(\rho(\tilde{u})(z - \tilde{z}) + \rho^*(\tilde{u}, \tilde{z})(u - \tilde{u}) \right) - \rho(\tilde{u})(\tilde{z}) + \mathcal{R}^{(3)}.$$

- **Components of the identity:**

- $\rho(\tilde{u})(\cdot) := -\mathcal{A}(\tilde{u})(\cdot).$
- $\rho^*(\tilde{u}, \tilde{z})(\cdot) := J'(\tilde{u}) - \mathcal{A}'(\tilde{u})(\cdot, \tilde{z}).$
- Remainder $\mathcal{R}^{(3)}$ contains higher-order derivatives.

DWR: Enriched Spaces

- **Enriched solve:** Compute

$$u_h^{(2)} \in U_h^{(2)}, \quad z_h^{(2)} \in V_h^{(2)},$$

with

$$U_h \subset U_h^{(2)} \subset U, \quad V_h \subset V_h^{(2)} \subset V.$$

- **Saturation assumption:**

$$|J(u) - J(u_h^{(2)})| \leq b_h |J(u) - J(\tilde{u})|, \quad 0 < b_h < 1.$$

- **Consequence:**

$$\begin{aligned}
 J(u) - J(\tilde{u}) &= \underbrace{J(u) - J(u_h^{(2)})}_{\text{small under assumption}} + \underbrace{J(u_h^{(2)}) - J(\tilde{u})} \\
 &= \frac{1}{2} \left(\rho(\tilde{u})(z_h^{(2)} - \tilde{z}) + \rho^*(\tilde{u}, \tilde{z})(u_h^{(2)} - \tilde{u}) \right) - \rho(\tilde{u})(\tilde{z}) + \mathcal{R}_h^{(3)}
 \end{aligned}$$

Goal-based Adaptivity

A DWR-driven adaptive workflow is

1. **Solve:** Compute the primal and adjoint solutions.
2. **Estimate (DWR):** Evaluate local error indicators η_K .
3. **Mark:** Select a subset of elements and time intervals (?) with the largest error contributions.
4. **Refine:** Perform local mesh refinement and adapt the time-step size accordingly.
5. **Termination condition:** Repeat until the estimated global goal error satisfies $|\eta| < \text{TOL}$, where $\eta = \sum_K \eta_K$.

Implementation Plan

- ▶ **Tools:** Firedrake for FEM assembly, Irksome for time discretisation, and Pyadjoint(?) for automatic adjoints.
- ▶ **Plan:** verify adjoints for time-dependent problems, implement linear DWR estimators manually, then automate the workflow.
- ▶ (*Aspirational Extension*) extend the approach to nonlinear problems.
- ▶ **Outcome:** a Firedrake-based tool for goal-oriented adaptivity in time-dependent PDEs.